

Nicolas Christianson

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Academic Appointments	Johns Hopkins University. Baltimore, MD Assistant Professor, Department of Computer Science	2026 –
	Stanford University. Stanford, CA Postdoctoral Scholar, Management Science and Engineering Mentors: <i>Ellen Vitercik and Ram Rajagopal</i> <i>Stanford Energy Postdoctoral Fellow</i>	2025 – 2026
Education	California Institute of Technology. Pasadena, CA Ph.D. in Computing and Mathematical Sciences. Advisors: <i>Adam Wierman and Steven Low</i> <i>NSF Graduate Research Fellow</i> <i>PIMCO Graduate Fellow in Data Science</i> <i>Resnick Sustainability Institute (RSI) Scholar</i>	2025
	Harvard College. Cambridge, MA A.B. <i>summa cum laude</i> in Applied Mathematics.	2020
Industry Collaborations	Microsoft Research <i>Developed new algorithms to reliably deploy machine learning to power grid contingency analysis in collaboration with the Microsoft Research Special Projects group; resulted in a paper at L4DC 2025 (conference publication 8).</i>	2023 – 2025
	Amazon Prime Video <i>Developed new algorithms for adaptive bitrate video streaming leveraging advancements in online optimization and learning. Yielded substantial improvements over the state-of-the-art and deployment to the Amazon Prime Video production environment, with results documented in a paper at SIGCOMM 2024 (conference publication 11).</i>	2023 – 2024
	Beyond Limits <i>Developed algorithms for robust and efficient operation of real-world electricity/steam cogeneration resources in grids with high renewables penetration. Wrote a manuscript documenting results (journal paper 4) and incorporated the system model into SustainGym, an open-source library of sustainability-related benchmarks for reinforcement learning, documented in a paper at NeurIPS 2023 (conference publication 16).</i>	2022 – 2025
Honors and Awards	ACM SIGEnergy Doctoral Dissertation Award	2026
	Stanford Energy Postdoctoral Fellowship	2025
	Ben P.C. Chou Doctoral Prize in Information Science and Technology	2025
	Demetriades-Tsafka-Kokkalis Prize in Renewable Energy Sources	2025
	PIMCO Graduate Fellowship in Data Science	2024
	NSF Graduate Research Fellowship	2021
	Phi Beta Kappa Junior 24	2019
	John Harvard Scholarship	2017, 2019
	Blair Research Fellowship (UPenn)	2018
	Detur Book Prize	2017

Working Papers

* indicates equal contribution

† indicates student I advised or coadvised

1. **Multi-Scale Datacenter Power Modulation**
Preprint
Akshay Sreekumar, **Nicolas Christianson**, Fiodar Kazhamiaka, Ram Rajagopal
2. **Online Smoothed Demand Management**
Journal version in preparation
Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Preliminary version accepted to ACM SIGMETRICS 2026 (conference publication 4)
3. **Online Conversion with Switching Costs: Robust and Learning-Augmented Algorithms**
Journal version under review
Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Preliminary version accepted to ACM SIGMETRICS/IFIP Performance 2024 (conference publication 14)
4. **Learning Dynamic Graphs, Too Slow**
Preprint
Andrei A. Klishin, **Nicolas H. Christianson**, Cynthia S.Q. Siew, Dani S. Bassett

Conference Publications

1. **Risk-Sensitive Peak-Aware Energy Scheduling: Competitive and Learning-Augmented Algorithms**
ACM e-Energy 2026
Finalist, Best Paper Award
Lukas Himmelreich†, **Nicolas Christianson**, Adam Wierman
2. **Competitive and Learned Online Algorithms for HVAC Demand Management in Thermal Networks**
ACM e-Energy 2026
Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
3. **Prediction-Specific Design of Learning-Augmented Algorithms**
ACM SIGMETRICS 2026
Sizhe Li†, **Nicolas Christianson**, Tongxin Li *Journal version: POMACS 2026 (journal publication 1)*
4. **Online Smoothed Demand Management**
ACM SIGMETRICS 2026
Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Journal version: In preparation (working paper 2)
5. **Improving EV Aggregate Flexibility with End-to-End Learning**
Learning for Dynamics & Control Conference (L4DC) 2026, accepted
Apoorva Thanvantri†, Christopher Yeh, **Nicolas Christianson**, Adam Wierman
6. **Conformal Risk Training: End-to-End Optimization of Conformal Risk Control**
39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)
Christopher Yeh, **Nicolas Christianson**, Adam Wierman, Yisong Yue

7. **Learning for Online Scheduling with Competitive Fairness Guarantees**
ACM e-Energy 2025
 Pengfei Li, **Nicolas Christianson**, Jianyi Yang, Adam Wierman, Shaolei Ren
8. **Fast and Reliable $N - k$ Contingency Screening with Input-Convex Neural Networks**
Learning for Dynamics & Control Conference (L4DC) 2025
Nicolas Christianson, Wenqi Cui, Steven Low, Weiwei Yang, Baosen Zhang
9. **Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints**
ACM SIGMETRICS 2025
 Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Journal version: POMACS 2025 (journal publication 3)
10. **Risk-Sensitive Online Algorithms**
37th Annual Conference on Learning Theory (COLT 2024)
Nicolas Christianson, Bo Sun, Steven Low, Adam Wierman
11. **SODA: An adaptive bitrate controller for consistent high-quality video streaming**
SIGCOMM 2024
 Tianyu Chen, Yiheng Lin, **Nicolas Christianson**, Zahaib Akhtar, Sharath Dharmaji, Mohammad Hajiesmaili, Adam Wierman, Ramesh K. Sitaraman
12. **Chasing Convex Functions with Long-term Constraints**
41st International Conference on Machine Learning (ICML 2024)
 Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
13. **Online Algorithms with Uncertainty-Quantified Predictions**
41st International Conference on Machine Learning (ICML 2024)
 Bo Sun, Jerry Huang[†], **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Raouf Boutaba
14. **Online Conversion with Switching Costs: Robust and Learning-Augmented Algorithms**
ACM SIGMETRICS/IFIP PERFORMANCE 2024
 Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Journal version: Under review (working paper 3)
15. **The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting**
ACM SIGMETRICS/IFIP PERFORMANCE 2024
 Adam Lechowicz, **Nicolas Christianson**, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Journal version: POMACS 2023 (journal publication 5)
16. **SustainGym: Reinforcement Learning Environments for Sustainable Energy Systems**
37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023), Datasets and Benchmarks Track
 Christopher Yeh, Victor Li, Rajeev Datta, Julio Arroyo, **Nicolas Christianson**, Chi Zhang, Yize Chen, Mohammad Mehdi Hosseini, Azarang Golmohammadi, Yuanyuan Shi, Yisong Yue, Adam Wierman

17. **Pricing Uncertainty in Stochastic Multi-Stage Electricity Markets**
62nd IEEE Conference on Decision and Control (CDC 2023)
 Lucien Werner*, **Nicolas Christianson***, Alessandro Zocca, Adam Wierman, Steven Low
18. **Optimal robustness-consistency tradeoffs for learning-augmented metrical task systems**
26th International Conference on Artificial Intelligence and Statistics (AISTATS 2023)
Nicolas Christianson, Junxuan Shen[†], Adam Wierman
19. **Smoothed Online Optimization with Unreliable Predictions**
ACM SIGMETRICS 2023
 Daan Rutten, **Nicolas Christianson**, Debankur Mukherjee, Adam Wierman
Journal version: POMACS 2023 (journal publication 6)
20. **Dispatch-aware planning for feasible power system operation**
22nd Power Systems Computation Conference (PSCC 2022)
Nicolas Christianson*, Lucien Werner*, Adam Wierman, Steven Low
Journal version: EPSR 2022 (journal publication 7)
21. **Chasing Convex Bodies and Functions with Black-Box Advice**
35th Annual Conference on Learning Theory (COLT 2022)
Nicolas Christianson, Tinashe Handina, Adam Wierman

Journal Publications

1. **Prediction-Specific Design of Learning-Augmented Algorithms**
Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 10, iss. 1, art. 18, pp. 1-57, 2026
 Sizhe Li[†], **Nicolas Christianson**, Tongxin Li
Also appeared at ACM SIGMETRICS '26 (conference publication 3)
2. **End-to-End Conformal Calibration for Optimization Under Uncertainty**
Transactions on Machine Learning Research, 2025
Honorable Mention, Best Poster Award at LANL Grid Science Winter School 2025
 Christopher Yeh*, **Nicolas Christianson***, Alan Wu, Adam Wierman, Yisong Yue
Preliminary versions appeared at Workshops on Tackling Climate Change with Machine Learning at ICLR '23 and NeurIPS '24 (workshop papers 2, 1)
3. **Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints**
Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 9, iss. 1, art. 8, pp. 1-49, 2025
 Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Also appeared at ACM SIGMETRICS '25 (conference publication 9)
4. **Robust Machine-Learned Algorithms for Efficient Grid Operation**
Environmental Data Science; vol. 4, art. e24
Nicolas Christianson, Christopher Yeh, Tongxin Li, Mehdi Hosseini, Mahdi Torabi Rad, Azarang Golmohammadi, Adam Wierman
Preliminary version appeared at NeurIPS '22 Workshop on Tackling Climate Change with Machine Learning (workshop paper 3)

5. **The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting**
Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 7, iss. 3, art. 45, pp. 1-32, 2023
Adam Lechowicz, **Nicolas Christianson**, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
Also appeared at *ACM SIGMETRICS/IFIP PERFORMANCE '24* (conference publication 15)
6. **Smoothed Online Optimization with Unreliable Predictions**
Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 7, iss. 1, art. 12, pp. 1-36, 2023
Daan Rutten, **Nicolas Christianson**, Debankur Mukherjee, Adam Wierman
Also appeared at *ACM SIGMETRICS/IFIP PERFORMANCE '24* (conference publication 19)
7. **Dispatch-aware planning for feasible power system operation**
Electric Power Systems Research; vol. 212: 108597, 2022
Nicolas Christianson*, Lucien Werner*, Adam Wierman, Steven Low
Also appeared at *PSCC '22* (conference publication 20)
8. **Optimizing the human learnability of abstract network representations**
Proceedings of the National Academy of Sciences; vol. 119, iss. 35: e2121338119, 2022
William Qian, Christopher W. Lynn, Andrei A. Klishin, Jennifer Stiso, **Nicolas H. Christianson**, Dani S. Bassett
9. **Architecture and evolution of semantic networks in mathematics texts**
Proceedings of the Royal Society A; vol. 476, iss. 2239: 20190741, 2020
Nicolas H. Christianson, Ann Sizemore Blevins, Dani S. Bassett
10. **Structural and Functional Influence of the Glycine-Rich Loop G³⁰²GGGY on the Catalytic Tyrosine of Histone Deacetylase 8**
Biochemistry; vol. 55, iss. 48: 6718-6729, 2016
Nicholas J. Porter, **Nicolas H. Christianson**, Christophe Decroos, David W. Christianson
11. **Biochemical and Structural Characterization of HDAC8 Mutants Associated with Cornelia de Lange Syndrome Spectrum Disorders**
Biochemistry; vol. 54, iss. 42: 6501-6513, 2015
Christophe Decroos, **Nicolas H. Christianson**, Laura E. Gullett, Christine M. Bowman, Karen E. Christianson, Matthew A. Deardorff, David W. Christianson

Workshop Papers

1. **End-to-End Conformal Calibration for Robust Grid-Scale Battery Storage Optimization**
Workshop on Tackling Climate Change with Machine Learning at NeurIPS 2024
Christopher Yeh*, **Nicolas Christianson***, Adam Wierman, Yisong Yue
Full version: TMLR (journal paper 2)
2. **Decision-Aware Uncertainty-Calibrated Deep Learning for Robust Energy System Operation**
Workshop on Tackling Climate Change with Machine Learning at ICLR 2023
Christopher Yeh, **Nicolas Christianson**, Steven Low, Adam Wierman, Yisong Yue
Full version: TMLR (journal paper 2)

3. **Robustifying Machine-Learned Algorithms for Efficient Grid Operation**
*Workshop on Tackling Climate Change with Machine Learning at **NeurIPS 2022***
Nicolas Christianson, Christopher Yeh, Tongxin Li, Mahdi Torabi Rad, Azarang Golmohammadi, Adam Wierman
Full version: Environmental Data Science (journal paper [4](#))

**Invited Talks
and Seminars**

End-to-End Learning for Uncertainty- and Risk-Aware Decision-Making
Stanford University - RAIN Seminar. February 2026

End-to-End Learning for Fast Contingency Screening
INFORMS Annual Meeting. October 2025

Reliable AI-Augmented Algorithms for Energy
ACM e-Energy '26 – Doctoral Dissertation Award Talk. June 2026
UC Berkeley – eMERGE Seminar. September 2025
Cornell University – ORIE Seminar. February 2025
MIT Sloan School of Management – OM Seminar. January 2025
Yale School of Management – Operations Seminar. January 2025
Johns Hopkins University – CS Seminar. December 2024
Columbia Business School – DRO Seminar. December 2024

Reliable ML-Augmented Algorithms for Energy and Sustainability
INFORMS Annual Meeting. October 2024
UMass Amherst – Systems and Sustainability Seminar. October 2024
Cornell ORIE – Young Researchers Workshop. October 2024
UC Berkeley – Energy Modeling, Analysis, & Control Group. September 2024

Learning-augmented algorithms for online optimization and beyond
Alberta Machine Intelligence Institute (Amii) AI Seminar. July 2024

Risk-Sensitive Online Algorithms
UMass Amherst – CS Theory Seminar. November 2025
INFORMS APS Conference. July 2025

Robust Machine-Learned Algorithms for Efficient Grid Operation
CAST Annual Program Review, Caltech. October 2023

Provable Guarantees on AI/ML for Metrical Task Systems and Classification
UMass Amherst – CS Theory Seminar. October 2023

Optimal Robustness-Consistency Tradeoffs for Learning-Augmented Metrical Task Systems
INFORMS Annual Meeting. October 2023

Chasing Convex Bodies and Functions with Black-Box Advice
Asilomar Conference on Systems and Signals. November 2022
Harvard University – Na Li’s Research Group. October 2022
UMass Amherst – Data Science Deep Dive Seminar. October 2022
INFORMS Annual Meeting. October 2022

Research Mentorship	Sizhe Li (CUHK Shenzhen B.S. '26)	2025 – 2026
	<i>Topic:</i> Learning-augmented online algorithms	
	<i>Next step:</i> Johns Hopkins CS PhD student (in my group)	
	Elizabeth Rogers (Harvey Mudd B.S. '26)	2025
	<i>Topic:</i> Health-aware optimal power flow	
	Apoorva Thanvantri (Caltech B.S. '26)	2025
	<i>Topic:</i> Improving electric vehicle aggregate flexibility with end-to-end learning	
	<i>Next step:</i> Harvard EE PhD student	
	Lukas Himmelreich (ETH Zurich M.Sc. '25)	2025
	<i>Topic:</i> Risk-sensitive algorithms for peak-aware energy scheduling	
	<i>Next step:</i> ML engineer at Julius Baer	
Teaching Experience	James Chen (Caltech B.S. '24)	2023 – 2024
	<i>Topic:</i> Learning-augmented online optimization with ramp constraints	
	<i>Next step:</i> MIT EECS PhD student	
	Junxuan (Helen) Shen (Caltech B.S. '24)	2022 – 2024
	<i>Topic:</i> Learning-augmented algorithms for multiserver convex function chasing	
	<i>Next step:</i> MIT EECS PhD student	
	Jerry Huang (Caltech B.S. '24)	2022 – 2023
	<i>Topic:</i> Online algorithms with uncertainty-quantified predictions	
	<i>Next step:</i> CMU CS PhD student	
Funding: Proposals and awards	Johns Hopkins University	
	EN.601.734: Topics in Machine Learning-Augmented Algorithm Design	Fall 2026
	California Institute of Technology	
	→ Teaching Assistant	
	CS 146: Control and Optimization of Networks	Spring 2024
	CS 42: Computer Science Education in K-14 Settings	Winter 2024
	CS 146: Control and Optimization of Networks	Winter 2023
	CS 42: Computer Science Education in K-14 Settings	Winter 2023
	Harvard College	
	→ Peer Tutor, Harvard Bureau of Study Counsel/Academic Resource Center	
	Math 25a: Theoretical Linear Algebra and Real Analysis I	Fall 2019
	CS 181: Machine Learning	Spring 2019
	APMTH 50: Introduction to Applied Mathematics	Spring 2019, Spring 2020
	Math 132: Differential Topology	Spring 2019
	APMTH 106: Applied Algebra	Fall 2018
	STAT 110: Introduction to Probability	Fall 2018
	CS 51: Intro to Computer Science II	Spring 2018, Spring 2019, Spring 2020
	Math 25b: Theoretical Linear Algebra and Real Analysis II	Spring 2018
	→ Course Assistant	
	Math Ma: Introduction to Functions and Calculus I	Fall 2017
	Assisted in preparation of two proposals (PI: Adam Wierman) through Caltech's Center for Autonomous Systems and Technology, funded by Beyond Limits in 2022 and 2023. \$230,000	
	NSF Graduate Research Fellowship, awarded 2021. \$138,000	

Professional Service	Workshop Organization	
	Learning-augmented Algorithms: Theory and Applications at ACM SIGMETRICS.	2024 – 2026
	Physics-Informed Learning for Optimization and Control of Sustainable Energy Systems at ACM e-Energy	2026
	AI Foundations for Power Grids at NeurIPS	2026
	TPC Membership	
	ACM SIGMETRICS	2027
	ACM e-Energy	2026, 2027
	GreenSys Workshop	2026
	Journal Reviewing	
	Journal of the ACM	2026
	IEEE Transactions on Automatic Control	2025, 2026
	IEEE Transactions on Control of Network Systems	2026
	Operations Research	2025
	IEEE/ACM Transactions on Networking	2023, 2024
	Conference and Workshop Reviewing	
	→ Conferences	
	Asilomar Conference on Signals, Systems, and Computers.	2022, 2024
	NeurIPS.	2025 (“Top Reviewer”), 2026
	IEEE Conference on Decision and Control (CDC).	2025
	Learning for Dynamics & Control Conference (L4DC).	2025, 2026
	ACM SIGMETRICS. (subreviewer)	2022, 2023, 2024
	ACM e-Energy. (subreviewer)	2023, 2024
	→ Workshops	
	NeurIPS Workshop on Computational Sustainability.	2023
	Grant Reviewing	
	Israel Science Foundation (ISF).	2026
	Internal University Service at Caltech	
	Student Member of Caltech CMS AI/ML Admissions Committee	2022 – 2025
	Caltech CMS Prelim Exam Preparation Coordinator	2021 – 2022
Outreach	Pasadena Public Schools. Science Night volunteer	2023 – 2025
	Designing and organizing hands-on CS activities for elementary school students for “Science Nights” at public schools in and around Pasadena.	
	Caltech Accountability Partners Program. PhD application mentor	2022 – 2025
	iSTEM Scholars. Summer research mentor	2021
	Project SHORT. PhD application mentor	2020 – 2025
Selected Work Experience	Microsoft Research. Redmond, WA	Summer 2023
	Research Intern Intern in the Special Projects group, developed reliable machine learning methods to accelerate contingency analysis in energy grids while ensuring provable guarantees on performance.	

KOACORE. Remote

Spring 2023

Machine Learning Lead and Consultant

Led development of proof-of-concept and product strategy for KOA-SUPPLY, an AI-driven healthcare supply chain marketplace which has since been acquired by Stead Impact Ventures and now operates as [Cato](#).

The Boston Consulting Group. Boston, MA

Summer 2019

Summer Associate

Partnered with a top-10 global biopharmaceutical company to optimize its supply and manufacturing networks, using data and digital-driven techniques to forecast production needs and increase efficiency.